

Smart EV Charging Station Optimizer

Problem Statement & Live Demo

ESP32 · MQTT · ThingsBoard · Edge AI

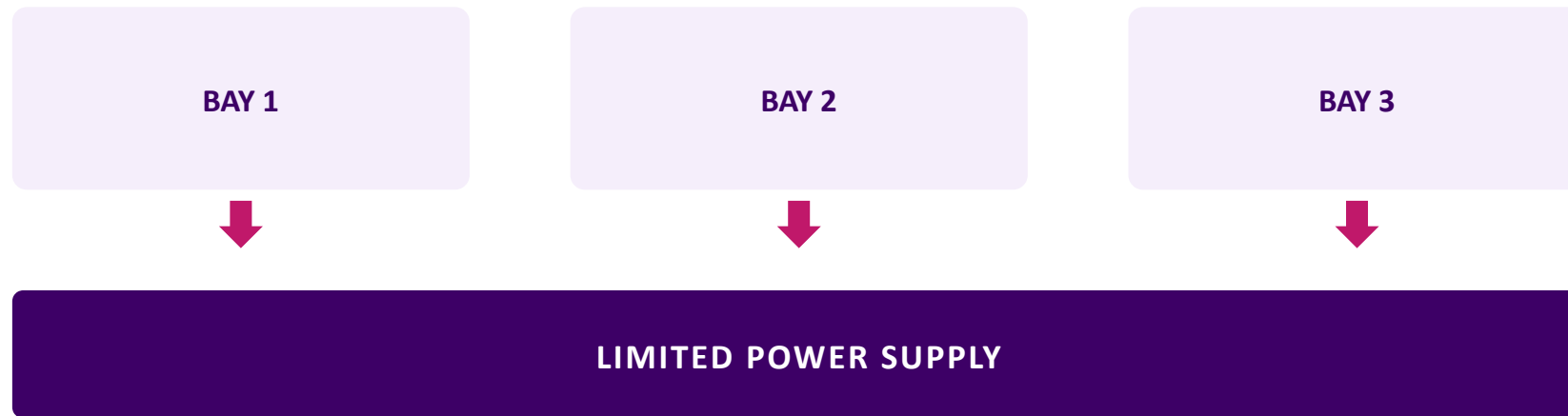
BAY 1

BAY 2

BAY 3

15-day build starts today

**Three EVs plug in to charge
at the same time.
What happens?**



Let's find out — together.

Can the supply handle it?



Load management · optimization · throttling · deferral

This is exactly the problem our 15-day build solves — not just charging an EV, but monitoring, predicting, and optimizing how every bay charges.

THE PROJECT

Smart EV Charging Station Optimizer

A multi-bay charging station that senses, predicts, and optimizes charging — in real time.

IoT

Edge AI

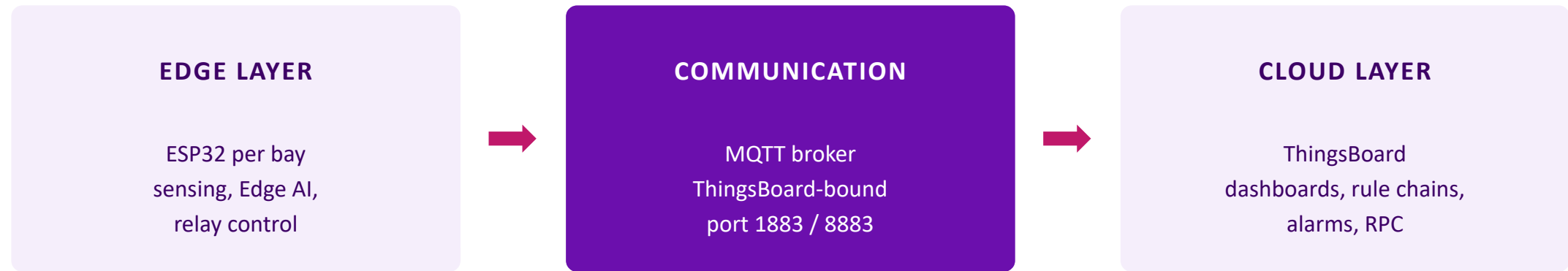
MQTT

ThingsBoard

Load Optimization

“We are going to simulate a multi-bay EV charging station. Each bay measures its electrical parameters. The ESP32 sends this information to the cloud. ThingsBoard monitors every bay, predicts upcoming demand, and decides how charging should be managed.”

Edge → Communication → Cloud



Edge AI runs locally on each ESP32 — low-latency decisions continue even if cloud connectivity is briefly lost. ThingsBoard adds cross-node aggregation, historical analytics, and operator alarms.

Over the next 30 days, we'll build these blocks one by one.

What's under the hood

"What technologies do you think are required?" — ask the class before revealing.

01

ESP32

the sensing + control board

02

Embedded C/C++

firmware running on ESP32

03

MQTT

lightweight IoT messaging

04

ThingsBoard

cloud dashboard & rules

05

Cloud Dashboard

live and historical views

06

IoT

connecting devices to cloud

07

Edge AI

on-device prediction

08

Optimization

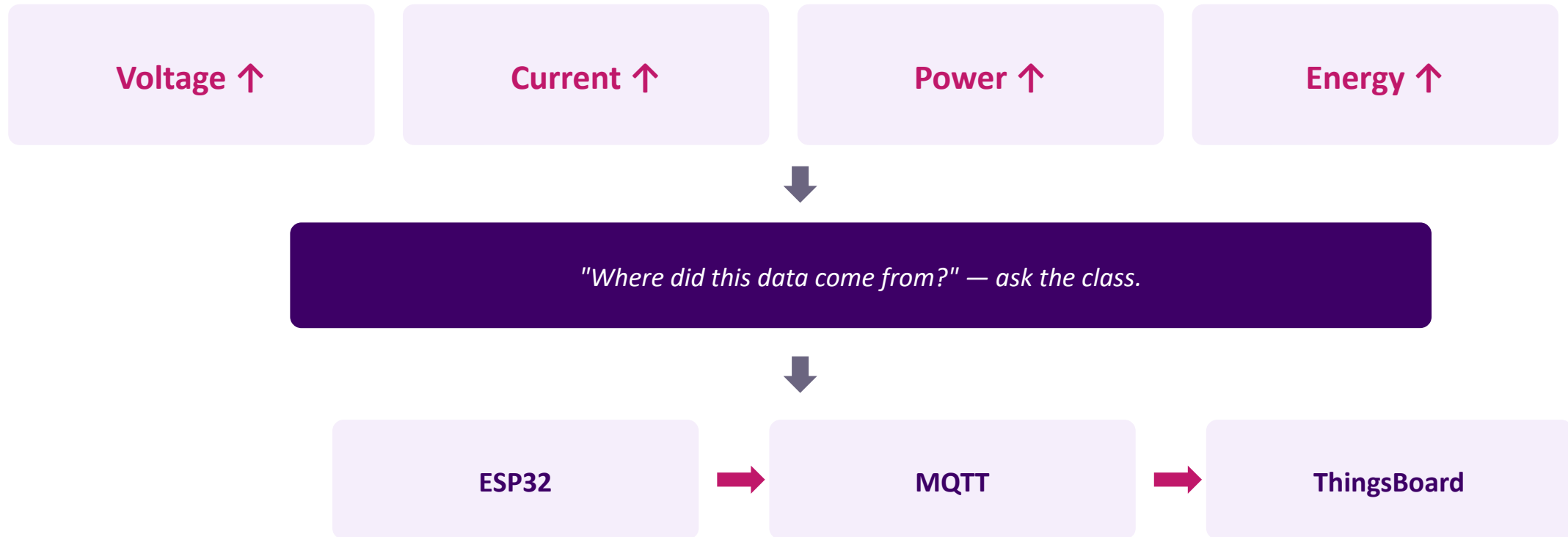
ALLOW / THROTTLE / DEFER

You don't need to know all of these today — by end, you'll have worked with every one.

Live Demo

Watch the complete system respond in real time.

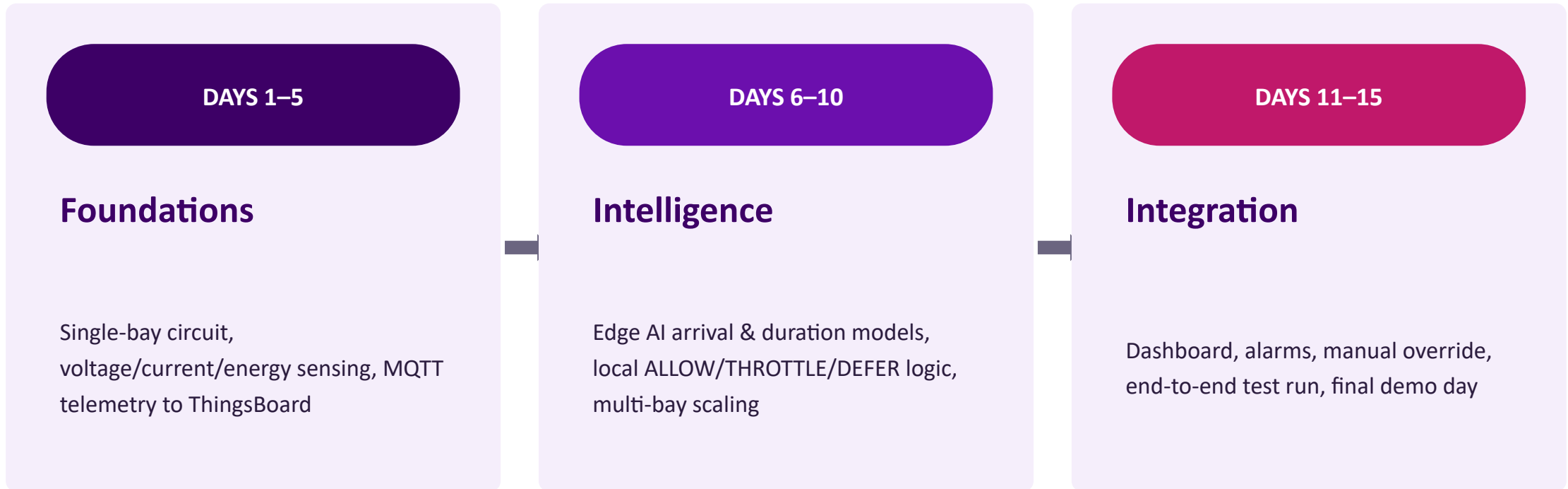
Bay 1 starts charging



What you will build

- A working multi-bay EV charging simulation where each bay independently senses, predicts, and decides
- A live ThingsBoard dashboard with real-time and historical data, trends, and active alarms
- A demonstrated reduction in peak power draw vs. an unmanaged baseline, via throttling and deferral
- A reusable, documented codebase: firmware + Edge AI training scripts + ThingsBoard configuration

15-day build plan



Class 1 challenge

1

Name the three layers in our system architecture.

2

Which protocol carries sensor data from the ESP32 to the cloud?

3

If demand is greater than supply, what should the system decide?

Think it over — we'll discuss the answers before Day 2 begins.

SEE YOU TOMORROW

Day 2: Circuit & Sensors

Bring your Wokwi / PICSimLab setup ready to go.

BAY 1

BAY 2

BAY 3